

PELLA, IA, USA

WMO: 720312

Lat: 41.400N

Lon: 92.946W

Elev: 270

StdP: 98.13

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04964

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.0	-17.4	-23.6	0.5	-19.1	-21.5	0.6	-16.7	12.7	-6.3	11.6	-6.1	3.0	320	0.496

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	33.9	24.3	32.4	23.5	30.6	22.7	25.8	30.8	24.9	29.8	23.9	28.9	3.7	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.6	20.2	28.9	23.0	18.4	27.5	22.4	17.7	26.8	81.3	30.8	76.9	29.6	72.9	28.7	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.6	9.0	8.0	-24.4	36.3	3.2	2.0	-26.7	37.7	-28.6	38.8	-30.4	39.9	-32.7	41.4		
(5)			-25.0	27.6	3.2	1.2	-27.2	28.5	-29.1	29.1	-30.9	29.8	-33.2	30.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.7	-4.6	-3.3	4.1	10.9	17.1	22.2	24.2	22.7	19.4	12.0	4.6	-1.6		
	DBStd	11.44	6.99	6.86	6.20	5.50	4.75	3.12	3.17	2.94	4.32	5.25	5.81	6.28		
	HDD10.0	1667	454	373	201	54	5	0	0	0	1	39	179	362		
	HDD18.3	3354	712	604	441	228	81	5	1	2	39	208	413	619		
	CDD10.0	1924	0	2	19	81	224	366	441	394	282	100	16	1		
	CDD18.3	572	0	0	1	6	42	121	184	138	70	10	0	0		
	CDH23.3	5343	0	0	9	78	372	1105	1859	1194	629	96	1	0		
	CDH26.7	1874	0	0	1	13	95	371	755	409	211	19	0	0		
(14) Wind	WSAvg	3.4	4.1	3.9	4.1	4.3	3.6	2.9	2.3	2.0	2.5	3.3	3.9	3.9		
(15) Precipitation	PrecAvg	925	26	35	55	95	119	140	115	114	88	63	47	31		
	PrecMax	1344	59	108	127	209	273	333	298	333	225	147	115	109		
	PrecMin	626	14	5	3	26	20	20	17	26	24	15	1	0		
	PrecStd	202	10	23	35	46	57	72	75	79	45	43	31	26		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.9	18.2	24.7	28.7	32.5	34.0	37.6	35.2	33.4	29.4	22.7	15.1		
		MCWB	7.4	11.3	16.8	17.7	21.7	24.7	24.1	22.5	23.0	20.0	14.5	11.2		
	2%	DB	9.1	12.9	20.4	25.9	29.0	32.6	34.7	32.8	31.3	26.1	18.7	11.5		
		MCWB	5.5	7.9	13.6	16.2	20.0	23.2	25.0	23.7	22.5	18.1	13.0	8.3		
	5%	DB	6.7	8.9	17.0	22.9	27.3	30.5	32.7	30.7	29.1	23.0	16.1	9.1		
		MCWB	4.1	5.4	11.4	14.5	18.6	22.1	24.2	22.8	21.3	16.8	11.3	6.6		
	10%	DB	4.1	5.9	13.6	20.4	25.3	28.6	30.6	28.7	27.2	20.9	13.4	6.4		
		MCWB	2.2	3.6	9.4	13.1	18.0	21.5	23.6	22.1	19.6	15.2	9.2	4.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.4	12.3	16.9	19.3	23.0	26.5	27.4	26.9	25.4	22.1	16.2	13.1		
		MCDB	11.0	17.3	23.0	26.5	29.0	31.4	32.5	32.2	30.6	26.4	19.7	14.7		
	2%	WB	6.1	8.7	14.8	17.4	21.6	24.9	26.0	25.3	23.7	20.2	14.3	9.5		
		MCDB	9.2	13.2	19.0	23.4	27.0	29.9	31.4	29.6	28.9	24.1	17.7	10.8		
	5%	WB	4.1	5.8	11.9	15.7	20.4	23.7	25.1	24.1	22.5	18.4	11.9	6.4		
		MCDB	6.4	8.7	16.3	21.0	25.6	28.7	30.3	28.3	27.1	21.4	15.3	8.7		
	10%	WB	2.3	3.4	9.4	14.3	19.3	22.6	24.2	23.0	21.4	15.8	9.7	4.2		
		MCDB	4.1	5.9	13.7	19.4	23.8	27.3	29.4	27.6	25.4	19.5	13.0	6.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.8	9.4	10.1	11.6	11.1	10.9	11.2	11.2	12.1	11.1	10.0	8.4		
		MCDBR	11.6	13.5	15.6	15.7	12.9	12.7	12.8	13.1	13.5	14.3	13.8	11.5		
	5% WB	MCWBR	8.6	9.1	9.3	8.4	6.2	5.9	5.2	5.7	5.7	7.6	8.8	8.7		
		MCWBR	11.2	13.2	13.1	13.2	11.2	11.1	10.8	10.8	11.4	11.1	12.7	10.6		
(40) Clear-Sky Solar Irradiance	taub	taub	0.288	0.311	0.333	0.385	0.411	0.421	0.438	0.419	0.385	0.341	0.300	0.289		
		taud	2.422	2.361	2.390	2.262	2.261	2.278	2.219	2.290	2.335	2.455	2.499	2.437		
	Ebn at Noon	Ebn at Noon	869	897	916	885	867	856	836	843	848	853	851	836		
		Edh at Noon	78	97	107	130	134	132	139	126	112	88	73	71		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.92	2.92	3.79	4.75	5.58	6.19	6.24	5.43	4.61	3.08	2.09	1.60		
		RadStd	0.18	0.21	0.35	0.35	0.42	0.62	0.46	0.33	0.34	0.35	0.18	0.15		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page